

ITA0607 – Machine Learning

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Unit I Analytical Assignments with SDGs

Explain waste collection optimization using Machine Learning.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
data = {
    'Bin_Fill_Level': [20,35,50,60,75,80,90,95,40,55,70,85,30,65,78],
    'Weight_kg': [5,8,10,12,15,18,20,22,7,11,14,19,6,13,17],
    'Days_Since_Last_Collection': [1,2,2,3,4,4,5,5,1,3,4,5,2,3,4],
    'Waste_Level': [
        'Low', 'Low', 'Medium', 'Medium', 'High',
        'High', 'High', 'High', 'Low', 'Medium',
        'High', 'High', 'Low', 'Medium', 'High'
    ]
}

df = pd.DataFrame(data)

df.head()
```

	Bin_Fill_Level	Weight_kg	Days_Since_Last_Collection	Waste_Level
0	20	5		Low
1	35	8		Low
2	50	10		Medium
3	60	12		Medium
4	75	15		High

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
print(df)

df.describe()
```

	Bin_Fill_Level	Weight_kg	Days_Since_Last_Collection	Waste_Level
0	20	5		1 Low
1	35	8		2 Low
2	50	10		2 Medium
3	60	12		3 Medium
4	75	15		4 High
5	80	18		4 High
6	90	20		5 High
7	95	22		5 High
8	40	7		1 Low
9	55	11		3 Medium
10	70	14		4 High
11	85	19		5 High
12	30	6		2 Low
13	65	13		3 Medium
14	78	17		4 High

	Bin_Fill_Level	Weight_kg	Days_Since_Last_Collection	
count	15.000000	15.000000		15.000000
mean	61.866667	13.133333		3.200000
std	23.024418	5.343443		1.373213
min	20.000000	5.000000		1.000000
25%	45.000000	9.000000		2.000000
50%	65.000000	13.000000		3.000000
75%	79.000000	17.500000		4.000000
max	95.000000	22.000000		5.000000

```
X = df[['Bin_Fill_Level', 'Weight_kg', 'Days_Since_Last_Collection']]
y = df['Waste_Level']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42
)
```

```
model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

print("Model Training Completed")
```

Model Training Completed

```
predictions = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, predictions))

print("\nClassification Report\n")
print(classification_report(y_test, predictions))
```

Accuracy: 1.0

Classification Report

	precision	recall	f1-score	support
High	1.00	1.00	1.00	2
Low	1.00	1.00	1.00	1
Medium	1.00	1.00	1.00	2
accuracy			1.00	5

macro avg	1.00	1.00	1.00	5
weighted avg	1.00	1.00	1.00	5

```
new_bin = [[88,19,5]]

prediction = model.predict(new_bin)

print("Predicted Waste Level:", prediction[0])
```

Predicted Waste Level: High
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not
warnings.warn(

```
level = prediction[0]

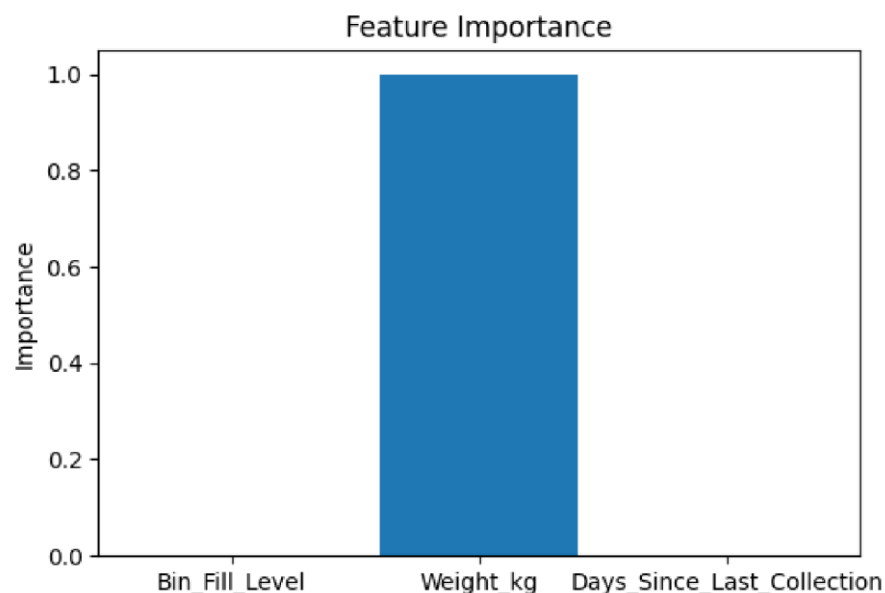
if level == "High":
    print("Priority: Collect Immediately")
elif level == "Medium":
    print("Priority: Schedule Collection Soon")
else:
    print("Priority: Collection Not Required Now")
```

Priority: Collect Immediately

```
importance = model.feature_importances_

features = X.columns

plt.figure(figsize=(6,4))
plt.bar(features, importance)
plt.title("Feature Importance")
plt.ylabel("Importance")
plt.show()
```



```
cm = confusion_matrix(y_test, predictions)

print("Confusion Matrix")
print(cm)
```

Confusion Matrix
[[2 0 0]
[0 1 0]
[0 0 2]]

```
fill = float(input("Enter Bin Fill Level (%): "))
weight = float(input("Enter Weight (kg): "))
days = float(input("Enter Days Since Last Collection: "))

sample = [[fill, weight, days]]

result = model.predict(sample)

print("\nPredicted Waste Level:", result[0])

if result[0] == "High":
    print("Collection Recommendation: Collect Immediately")
elif result[0] == "Medium":
    print("Collection Recommendation: Schedule Collection Soon")
else:
    print("Collection Recommendation: No Collection Needed")
```

```
Enter Bin Fill Level (%): 34
Enter Weight (kg): 50
Enter Days Since Last Collection: 4
```

```
Predicted Waste Level: High
Collection Recommendation: Collect Immediately
```